Probability Problems & Solutions

A worked collection of probability interview problems, with full explanations

18 probability problems, each with a step-by-step explanation of the reasoning behind the solution.

Problem 1 — Same-Side Probability

Statement

We have two coins. One is fair; the other is biased with P(heads) = 3/4. We select a coin at random and flip it twice. What is the probability that both flips show the same side?

Solution

"Both flips the same side" means two heads or two tails. Since we don't know which coin we picked, we compute the probability for each coin separately and then combine them, because each coin is equally likely to be selected.

For the biased coin: P(both same) = P(HH) + P(TT) = (3/4)(3/4) + (1/4)(1/4) = 9/16 + 1/16 = 10/16 = 0.625.

For the fair coin: P(both same) = (1/2)(1/2) + (1/2)(1/2) = 1/2.

By the law of total probability, weighting each coin by its 1/2 chance of selection:

$$P(\text{same}) = \tfrac{1}{2} \times 0.625 + \tfrac{1}{2} \times 0.5 = 0.3125 + 0.25 = 0.5625.$$

Answer: 0.5625

Problem 2 — Expected Rounds to Form Teams

Statement

Six people split into two equal teams. Each round, every person shows a hand face-up or face-down (50/50, independent). If exactly three are face-up, they split into teams. What is the expected number of rounds before they split?

Solution

Since they want random teams and no other information is given, we assume each person is equally likely (50/50) to show face-up or face-down. So each person's hand is a Bernoulli trial with success probability p = 0.5. A valid split happens when exactly 3 hands are face-up, since that forces the remaining 3 to be face-down.

Let F be the number of face-up hands. As a sum of independent Bernoulli trials, F follows a Binomial(6, 0.5) distribution, so the probability of a successful round is:

$$P(F = 3) = C(6,3)(1/2)^3(1/2)^3 = 20/64 = 0.3125.$$

Now let R be the number of rounds until the first successful split. R follows a geometric distribution, because it counts trials up to and including the first success. For a geometric random variable with success probability p, E[R] = 1/p. Therefore:

$$E[R] = 1 / 0.3125 = 3.2 \text{ rounds.}$$

Answer: 3.2 rounds

Problem 3 — Ad Raters

Statement

People rate ads. 80% are careful and rate an ad good with probability 0.6 (bad with 0.4); 20% are lazy and rate every ad good. (a) With 100 raters each rating one ad, what is the expected number of good ads? (b) With 1 rater rating 100 ads, what is the expected number? (c) Given one ad was rated bad, what is the probability the rater was lazy?

Solution

Let G be the event "rated good" and C the event "careful rater". A careful rater (80%) rates good with probability 0.6; a lazy rater (20%) always rates good. By the law of total probability:

$$P(G) = P(G|C)P(C) + P(G|C\blacksquare)P(C\blacksquare) = 0.6 \times 0.8 + 1 \times 0.2 = 0.68.$$

(a) With 100 independent raters each rating one ad, the number of good ratings follows a Binomial(100, 0.68). The expected value of a binomial is np, so the expected number of good ads = 100 × 0.68 = 68.

(b) With a single rater rating 100 ads, we condition on that rater's type:

$$E[\text{good}] = P(C)(100 \times 0.6) + P(C\blacksquare)(100 \times 1) = 0.8 \times 60 + 0.2 \times 100 = 48 + 20 = 68.$$

This matches part (a): mathematically it makes no difference whether 100 raters each rate one ad or one rater rates 100 ads.

(c) A lazy rater never rates an ad bad, so seeing a "bad" rating rules out a lazy rater entirely. Hence P(bad|lazy) = 0, which gives P(lazy|bad) = 0.

Answer: (a) 68 (b) 68 (c) 0

Problem 4 — First to Roll a Six

Statement

Amy and Brad take turns rolling a fair die; first to roll a 6 wins. Amy rolls first. What is the probability Amy wins?

Solution

Amy wins either on her very first roll, or only after both players miss a complete round and then she rolls a 6, and so on. Each miss has probability 5/6, and a 6 has probability 1/6. Summing over all the rounds in which Amy could win:

$$P(\text{Amy wins}) = (1/6) + (5/6)^2(1/6) + (5/6)\blacksquare(1/6) + \ldots = (1/6)[1 + (5/6)^2 + (5/6)\blacksquare + \ldots].$$

The bracket is a geometric series with common ratio $(5/6)^2$, which sums to $1 / (1 - (5/6)^2)$. Therefore:

$$P(\text{Amy wins}) = (1/6) / (1 - 25/36) = (1/6) / (11/36) = (1/6)(36/11) = 6/11.$$

Answer: 6/11 ≈ 0.545

Problem 5 — Jar of Coins

Statement

A jar holds 1000 coins: 999 fair and 1 double-headed. You pick a coin at random and toss it 10 times, getting 10 heads. (a) Probability the coin is double-headed? (b) Probability the next toss is also heads?

Solution

Let F be "fair coin" and D be "double-headed coin", with P(F) = 999/1000 = 0.999 and P(D) = 1/1000 = 0.001. A double-headed coin always lands heads, so P(10H|D) = 1, while a fair coin gives P(10H|F) = $(1/2)^{1\blacksquare}$. By the law of total probability:

$$P(10H) = P(10H|D)P(D) + P(10H|F)P(F) = 1 \times 0.001 + (1/2)^{1\blacksquare} \times 0.999 \approx 0.0019756.$$

(a) By Bayes' rule:

$$P(D|10H) = P(10H|D)P(D) / P(10H) = (1 \times 0.001) / 0.0019756 \approx 0.506.$$

(b) Whether the next toss is heads depends on which coin we are holding, so we use the updated (posterior) belief from part (a):

$$P(\text{next heads}) = P(H|D)P(D|10H) + P(H|F)P(F|10H) = 1 \times 0.506 + 0.5 \times (1 - 0.506) \approx 0.753.$$

Here the next toss is treated as dependent on the previous 10 tosses, through the posterior probability that the coin is double-headed.

Answer: (a) ≈ 0.506 (b) ≈ 0.753

Problem 6 — Three Increasing Cards

Statement

A deck of 500 cards numbered 1–500 is shuffled. You draw three cards one at a time. What is the probability each card is larger than the previous one?

Solution

Treat this as a sample-space problem and ignore the distracting population size. When you draw three distinct cards, there is effectively a lowest, a middle, and a highest card. To see the answer clearly, suppose the three drawn numbers are 1, 2, and 3. The "winning" outcome — each card larger than the last — is the single ordering (1, 2, 3). But the full set of possible orderings of three items is:

(3,2,1), (3,1,2), (2,1,3), (2,3,1), (1,3,2), (1,2,3).

That is 3! = 6 equally likely orderings, and exactly one of them is strictly increasing. So the probability is 1/6. The key insight is not to be distracted by the size of the population — it is irrelevant when you only care about the relative order within the sample.

Answer: 1/6

Problem 7 — Second Card Not an Ace

Statement

You draw two cards one at a time from a shuffled 52-card deck. What is the probability the second card is not an ace?

Solution

There are two disjoint ways for the second card to be a non-ace. Either the first card was an ace — probability 4/52 — leaving 48 non-aces among the remaining 51 cards; or the first card was a non-ace — probability 48/52 — leaving 47 non-aces among 51. Adding these cases:

$$P = (4/52)(48/51) + (48/52)(47/51) = [48(4 + 47)] / (52 \times 51) = (48 \times 51) / (52 \times 51) = 48/52 = 12/13 \approx 0.923.$$

Answer: 12/13 ≈ 0.923

Problem 8 — Four-Person Elevator

Statement

Four people ride an elevator in a building with 5 upper floors. Each picks a floor uniformly at random and independently. What is the probability that no two get off on the same floor?

Solution

"No two get off on the same floor" means all four people choose distinct floors. The number of ways to assign 5 floors to 4 people with no repetition is 5 × 4 × 3 × 2. The total number of unrestricted assignments is 5■, since each of the 4 people independently picks one of 5 floors. Therefore:

P(all distinct) = (5 × 4 × 3 × 2) / 5■ = 120 / 625 = 0.192.

Answer: 0.192

Problem 9 — Is the Dice Game Worth Playing?

Statement

You roll two fair dice. If the sum is 7 you win $21, but each roll costs $10. Is the game worth playing?

Solution

Let D■ and D■ be the two dice. The sum equals 7 in six ways: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), so P(sum = 7) = 6/36 = 1/6. Let P be your profit per roll. You either win, netting $21 – $10 = $11 with probability 1/6, or lose your $10 stake with probability 5/6. So the profit takes the values {+11, –10}, and its expected value is:

$$E[P] = 11 \times (1/6) - 10 \times (5/6) = (11 - 50)/6 = -39/6 \approx -6.5.$$

Because the expected profit is negative, you lose money on average, so the game is not worth playing.

Answer: E[P] ≈ –\$6.5 → not worth playing

Problem 10 — Five Heads out of Six

Statement

A biased coin shows heads 30% of the time. What is the probability of exactly 5 heads in 6 tosses?

Solution

The number of heads in 6 independent tosses with P(heads) = 0.3 follows a Binomial(6, 0.3) distribution. The probability of exactly 5 heads chooses which 5 of the 6 tosses are heads and multiplies by the corresponding probabilities:

P(5 heads) = C(6,5)(0.3)■$(0.7)^1$ = 6 × 0.00243 × 0.7 ≈ 0.0102.

Answer: ≈ 0.0102

Problem 11 — Three Zebras

Statement

Three zebras sit at the corners of an equilateral triangle. Each independently picks a direction (clockwise or counter-clockwise) along the triangle's edges and runs. What is the probability none collide?

Solution

The zebras avoid all collisions only if they all run the same way around the triangle — either all clockwise or all counter-clockwise. Any mix of directions causes at least one collision. Each zebra independently picks a direction with probability 1/2, so:

P(all clockwise) = $(1/2)^3$ = 1/8, and P(all counter-clockwise) = $(1/2)^3$ = 1/8.

P(no collision) = 1/8 + 1/8 = 1/4.

Answer: 1/4

Problem 12 — Raining in Seattle [CORRECTED]

Statement

You call 3 friends in Seattle and independently ask each if it's raining. Each tells the truth with probability 2/3 and lies with probability 1/3. All three say "Yes, it's raining." What is the probability that it is actually raining?

Solution

We want P(rain | all three say "Yes"). This is a Bayes' rule problem and requires a prior probability of rain, which we call p. If it is raining, all three say "Yes" only when all tell the truth, so P(3×Yes | rain) = $(2/3)^3$ = 8/27. If it is not raining, all three say "Yes" only when all lie, so P(3×Yes | no rain) = $(1/3)^3$ =

1/27. By Bayes' rule:

$$P(\text{rain} \mid 3\times\text{Yes}) = (8/27)p \,/\, [\,(8/27)p + (1/27)(1 - p)\,] = 8p \,/\, (7p + 1).$$

With an uninformed prior $p = 1/2$, this gives $8(1/2) / (7(1/2) + 1) = 4 / 4.5 = 8/9 \approx 0.889$.

> ***Correction note.*** *A common shortcut computes $1 - (1/3)^3 = 26/27$, but that is the probability that at least one friend is telling the truth — not the probability that it is raining. The correct approach uses Bayes' rule and requires a prior probability of rain; with an uninformed prior of 1/2 the answer is 8/9.*

Answer: 8/9 ≈ 0.889

Problem 13 — Netflix Lazy Raters

Statement

80% of raters are careful and rate 60% of movies good (40% bad); 20% are lazy and rate 100% of movies good. All raters rate equally many movies. What is the probability a movie is rated good?

Solution

Let G be the event that a movie is rated good. We condition on the rater's type. Careful raters make up 80% of raters and rate good with probability 0.6; lazy raters make up 20% and rate good with probability 1. By the law of total probability:

$$P(G) = P(G|\text{careful})P(\text{careful}) + P(G|\text{lazy})P(\text{lazy}) = 0.6 \times 0.8 + 1 \times 0.2 = 0.68.$$

Answer: 0.68

Problem 14 — Secret Wins

Statement

100 students each toss a coin. Heads on the first toss is a real win. On tails, they toss again: heads on the second toss means they lie and claim a win; tails means they admit a loss. If 30 students claim they won, how many actually won?

Solution

Let W be the number of students who actually won. Since 70 students did not claim a win, we know $W \leq 30$. Now consider the probability a student truly won given that they claimed a win. With a fair coin, a genuine win (heads on the first toss) has probability 0.5. But claiming a win has probability 0.75, because it also includes the $0.25 = 0.5 \times 0.5$ chance of getting tails then heads and lying about it. Therefore:

$$P(\text{won} \mid \text{claimed win}) = P(\text{won}) / P(\text{claimed win}) = 0.5 / 0.75 = 2/3.$$

Everyone who actually won also claimed a win, so among the 30 students who claimed a win, the number who truly won follows a Binomial(30, 2/3). Its expected value is np:

$$E[W] = 30 \times 2/3 = 20.$$

Answer: 20 students

Problem 15 — Marble Buckets [PART 2 CORRECTED]

Statement

Bucket 1 has 30 red and 10 black marbles; Bucket 2 has 20 red and 20 black. A friend secretly picks one bucket and draws a marble that turns out red. (a) Probability it came from Bucket 1? (b) She returns it and draws again from the SAME bucket (with replacement); both draws are red. Probability both came from Bucket 1?

Solution

Bucket 1 has 30 red and 10 black (40 total); Bucket 2 has 20 red and 20 black (40 total). Each bucket is equally likely to be chosen, so P(B1) = P(B2) = 1/2. The chance of drawing red from each is P(red|B1) = 30/40 = 3/4 and P(red|B2) = 20/40 = 1/2, giving an overall P(red) = ½(3/4) + ½(1/2) = 3/8 + 1/4 = 5/8.

(a) By Bayes' rule:

P(B1|red) = P(red|B1)P(B1) / P(red) = (3/4 × 1/2) / (5/8) = (3/8) / (5/8) = 3/5.

(b) She draws twice, with replacement, from the same unknown bucket, and both are red. Because both draws come from the same bucket, we must condition on both reds jointly rather than treating them separately:

P(RR|B1) = $(3/4)^2$ = 9/16, and P(RR|B2) = $(1/2)^2$ = 1/4, giving P(RR) = ½(9/16) + ½(1/4) = 9/32 + 4/32 = 13/32.

P(B1|RR) = P(RR|B1)P(B1) / P(RR) = (9/16 × 1/2) / (13/32) = (9/32) / (13/32) = 9/13.

> ***Correction note.*** *Because both marbles come from the same unknown bucket, the two reds must be conditioned on jointly. Treating them as independent posteriors — $(3/5)^2$ = 9/25 — is incorrect; the correct probability is 9/13.*

Answer: (a) 3/5 (b) 9/13 ≈ 0.692

Problem 16 — Gambler's Ruin

Statement

You start with $30. Each fair coin flip wins $1 (heads) or loses $1 (tails). You play until you reach $0 or $100. What is the probability you reach $100?

Solution

This is the classic Gambler's Ruin problem, modeled as a random walk. Starting at $30, at each step the gambler moves up or down by one dollar — from $k he can go only to $(k+1) or $(k–1). Each step is equally likely (probability 0.5) and independent of the past, so the walk is symmetric and fair. Because wins and losses balance out on average, the expected amount of money stays at the starting value, so E[X] = 30.

The walk must eventually stop at one of its two endpoints, \$0 or \$100. Let r be the probability of reaching \$100; then 1 – r is the probability of reaching \$0. Using the definition of expected value over these two outcomes:

$$E[X] = 100 \times r + 0 \times (1 - r) = 100r.$$

Since E[X] = 30, we have 100r = 30, so r = 0.3.

Answer: 30%

Problem 17 — Fake Review Detection

Statement

98% of reviews are legitimate and 2% are fake. If a review is fake, the algorithm flags it as fake 95% of the time. If a review is legitimate, the algorithm correctly identifies it as legitimate 90% of the time. If the algorithm flags a review as fake, what is the probability it is actually fake?

Solution

We want P(fake | flagged as fake). The base rates are P(fake) = 0.02 and P(legit) = 0.98. A fake review is flagged with probability 0.95. A legitimate review is wrongly flagged with probability 1 – 0.90 = 0.10. By Bayes' rule:

$$P(\text{fake} \mid \text{flagged}) = P(\text{flag}|\text{fake})P(\text{fake}) / [\ P(\text{flag}|\text{fake})P(\text{fake}) + P(\text{flag}|\text{legit})P(\text{legit})\ ] = (0.95 \times 0.02) / (0.95 \times 0.02 + 0.10 \times 0.98) = 0.019 / (0.019 + 0.098) = 0.019 / 0.117 \approx 0.162.$$

Notice how the large number of legitimate reviews drags the probability down to only about 16%, even though the detector is fairly accurate — a classic base-rate effect.

Answer: ≈ 0.162 (about 16.2%)

Problem 18 — Biased Coin Among Many

Statement

A bag has 100 coins; 1 is double-headed (both sides heads), the rest are fair. You pick a coin at random and toss it 3 times, getting heads all three times. What is the probability the coin is the biased one?

Solution

Let A be "the chosen coin is the double-headed one" and B be "3 heads in 3 tosses". The priors are P(A) = 1/100 and P(A■) = 99/100. The biased coin always lands heads, so P(B|A) = 1; a fair coin gives P(B|A■) = $(1/2)^3$ = 1/8. By Bayes' rule:

P(A|B) = P(B|A)P(A) / [P(B|A)P(A) + P(B|A■)P(A■)] = (1 × 1/100) / (1 × 1/100 + 1/8 × 99/100) = (1/100) / [(1/100)(1 + 99/8)] = 1 / (107/8) = 8/107.

Answer: 8/107 ≈ 0.075